



Document Profile

PHP Physical Environment Filing (Archive

1996-2011)

Status:	Final	Drawer:	Toxic Substances
Date:	02/08/2004	Folder:	Organochlorines\Serum Dioxin
Title:	Re: CONFIDENTIAL: Serum dioxin - David Phillips' response		Study\Phase 2\ESR
Author:	Peter Abernethy	File Location:	
Document Type:	Email	Unit:	
Summary:		Maintainer(s):	Annie Coughlan
Knowledge Content:			

Thanks Deborah,

It seems the issue ESR is trying to resolve is:

- (i) undertaking to participants that their results would be provided in two months or less
- (ii) a requirement that any change in protocol requires the ethics committee to be informed
- (iii) ESR needs to tell participants (and new participants) something as part of round two and does not wish to be misleading or withhold information.

Our solution is:

- (i) all participants told the study extended to allow us to have better information, results now expected to be at least 3 months away (phone call from ESR contracted nurse, followed by letter, would be my preference)
- (ii) ESR notify ethics committee that there has been some slight changes which are being managed through and offer briefing for them (with Deborah) if required
- (iii) new participants given same info as existing participants, results likely to be 3 months away.

Resolving differences:

- (i) may have to have something up our back sleeve for this (like referring OTAG advice to ethics committee and get their advice)?

Peter

Peter Abernethy
Manager
Communications
Government & Sector Relations
Corporate & Information Directorate
Ministry of Health
DDI: s 9(2)(g)(ii)

<http://www.moh.govt.nz>
mailto:peter_abernethy@moh.govt.nz

Deborah Read/MOH

Deborah Read/MOH

02/08/04 10:51 AM

To Sally Gilbert/MOH@MOH

cc Annie Coughlan/MOH@MOH, graeme.gillespie@moh.govt.nz,
Peter Abernethy/MOH@MOH

Subject Re: CONFIDENTIAL: Serum dioxin - David Phillips' response



Sally

The application to the Ethics Committee does not state when the participants will receive their results and anonymised aggregate group results other than once the study has been completed. Note that study completion includes peer review and to date ESR have not had the analysis reviewed (the model was reviewed before selection of participants began). OTAG also recommended on 7 July additional analysis as the report presented to OTAG was preliminary only.

The EC application does state that "if the protocol for this research changes in any way" the principal investigator (David Phillips) must inform the Ethics Committee. I do not know whether ESR did notify the Committee or not that the blood collection was going to occur in 2 stages and the reason for that. The informed consent form signed at the time of blood collection states "I will be given feedback on my individual levels and results [the results may take 2 months]". At the time that the blood was taken in late February it was already known that a second lot of bloods was likely to be taken depending on the results of the first stage - hence this time period should have been stated as longer. Also even if a second round of bloods was not going to eventuate 2 months was insufficient time given that the results once received by ESR had to be analysed and peer reviewed.

The predominant reason the study proceeded with a dual stage blood collection instead of 50 sampled at one time arose from OTAG advice in late 2003, that given uncertainty with the model it would be better to first choose a small potentially highly exposed group from whom we were most likely to see a statistically significant elevation in TCDD results above background if one existed, and then use those results to better inform the selection of the remaining individuals. The result of that was that 24 blood samples were taken in late February. Delays were then experienced with laboratory analysis and the results were received by ESR in 2 batches (final results were not received from Axys until mid June).

OTAG teleconference minutes of 17 May which discussed the results of the first 10 of 24 samples sent to Axys state: "exposure event(s) but don't know when, however next 14 samples might help pinpoint the time frames for exposure a bit better"

The 14 results that we hadn't received at that time along with the 10 discussed by OTAG on 7 July have NOT answered the question about when exposure occurred. In fact the results as a whole raise doubts about the assumption that the period of peak exposure was before implementation of the Clean Air Act 1972 in 1974 and raise the possibility that the 1986 bursting disk incident may have been more significant than currently thought on the basis of the information available from that time.

OTAG considered it would be beneficial in terms of communicating risk to the affected community if we could identify the period of exposure more precisely. It was therefore proposed that those selected to make up the total of 50 (ie the further 25) would include some individuals with residence that did not include the 1960s and early 1970s. The question of timing of release of results to the participants was discussed and the consensus was that the individual results would be best released to all participants when the study was completed. This delay would enable better information to be given to the participants as to when their peak exposure was most likely to have occurred and possibly also the extent of their exposure. These people do however need to be informed that receipt of their results has been delayed as the study has not yet been completed and given an indication of when that it is likely to be. If any individual indicated they were unwilling to wait he/she could be given their result but would not receive the results of the aggregate group at that stage. Hence they would not know how their exposure compared to others in Paritutu unless more than one individual couldn't wait and they communicated among themselves. This is possible given the small size of the community and in one case where more than one household member had blood taken.

If the results are released now and the proposed total 50 blood samples is not completed the only statement about exposure that can be made is that the residents of Paritutu were exposed to TCDD

significantly above background levels during the period 1962-1987 (i.e. the period that former IWD was manufacturing 2,4,5-T). It is difficult to know how many people lived there for any of those 25 years other than it is in the order of thousands (???10,000). The number of people exposed will be less if the exposure period can be better identified. Given that the health significance of dioxin can only be described on a population basis (i.e. the clinical significance of an elevated TCDD result for an individual is unknown based on current scientific knowledge), and then only in a limited way, if we have a more precisely identified exposure period there will be fewer anxious people. On a population basis the results of epidemiological studies show small increases above background e.g. all cancers combined. If the Ministry of Health decides to set up an exposure register as an outcome of this study, whereby the health of the population is monitored over time based on hospitalisation, mortality and cancer registration data, if this small risk also exists in Paritutu it will not be able to be demonstrated if the population under surveillance includes individuals who were not exposed. Note there are other issues about setting up an exposure register given difficulties in identifying all who lived there and verifying the period of residence. Some residents would also have had occupational exposure (these residents were not included in the serum dioxin study).

Unfortunately I will not be able to attend the mtg with ESR on Thursday as I have meetings arranged elsewhere in one of my other roles.

Deborah

Deborah Read
Contractor
Environmental Health
Public Health Programmes
Public Health Directorate
Ministry of Health
DDI: § 9(2)(g)(ii) (Mondays only)

<http://www.moh.govt.nz>
mailto:Deborah_Read@moh.govt.nz

Sally Gilbert/MOH

Sally Gilbert/MOH

30/07/04 16:44

To graeme.gillespie@moh.govt.nz, Annie Coughlan/MOH@MOH,
Deborah Read/MOH@MOH, Peter Abernethy/MOH@MOH

cc

Subject CONFIDENTIAL: Serum dioxin - David Phillips' response

Hi everyone

David has just phoned regarding my email to him (below). He advises that he has obtained legal (ESR's in-house advisor) and ethical advice on this issue and he has briefed John Hay, and discussed the issue internally. David advises that ESR has a strong divergence of view from the Ministry. David initially indicated this divergence may be fundamental and insurmountable but during our conversation did agree that a meeting would be good to discuss everything and see what could be resolved between our agencies.

David's concern now seems to be, in addition to their ethical approval, that if ESR ask for more blood samples (for part 2 of the blood collection) people will want to know what happened to the results from the first round of collection. David is of the view that the peer reviews of the model show the model is

robust and there is sufficient information now to interpret the results etc for the participants.

David has requested the minutes from the OTAG meeting to be clear about OTAG's advice. He will be writing to outline his concerns and thinks a meeting would be a useful idea. I suggested piggy-backing this on the routine meeting between Bill/Fiona and Graeme next week and will confirm with David and Jeff if this suits Graeme. David will be bringing ESR's legal advisor so we may also want to invite Grant - and Peter - and maybe an ethicist (if we have one).

If David and Jeff (**AND OTAG!!!**) are comfortable that the model is robust, explains the blood results from part 1, and provides sufficient information to identify exposures (when they occurred, where they may have come from, what the exposures may have been, what this means for participants and the wider community) then it may be that we no longer need to proceed with part 2 and can proceed instead with the full release of the report. In my free and frank opinion, we could then cancel the rest of the contract with ESR and use the funding that is no longer required for part of the analysis to address concerns and put measures in place for the community. Mind you, again in my free and frank opinion, I am a little sceptical over this emerging information from ESR as David had omitted part 2 of the blood collection from the project specification and funding as far back as February this year, well before any results of peer review of the model and we need to be sure there is no underlying agenda and our actions are scientifically robust and would withstand objective scrutiny!!

Cheers - Sally

Sally Gilbert
Team Leader (Environment Team)
Environmental Health
Public Health Programmes
Public Health Directorate
Ministry of Health
DDI: s 9(2)(g)(ii)
Fax: 04 495 4401

<http://www.moh.govt.nz>
mailto:sally_gilbert@moh.govt.nz

----- Forwarded by Sally Gilbert/MOH on 30/07/04 04:28 pm -----

Sally Gilbert/MOH

30/07/04 04:22 pm

To david.phillips@esr.cri.nz

cc jeff.fowles@esr.cri.nz

Subject Serum dioxin

Dear David

It was good to catch up with you at the meeting at KSC on Wednesday. I have been mulling over our chat about the serum dioxin study and your concern that you may not be meeting the requirements of your ethical approval.

I wonder if this has arisen because we have broken the collection and analysis of bloods into two groups, when originally we envisaged that the collection and analysis would all be done in one group? I am not sure how your ethical approval is worded (we would be happy to review it if you are able to send it to us) but I can't imagine there is any expectation that results are provided to participants prior to the completion of the study. We certainly intend to meet the intent of the approval, which is to provide the results of the

study to everyone in a very timely manner!

You may recall at OTAG that this very issue was debated and the advice to the Ministry was that the study be completed before any results are given to participants. The reasoning was that the public health principle of 'first do no harm' was regarded as taking precedence and this would surely be consistent with any ethical Committee's requirements? I am very concerned that premature release of results create unwarranted concern among participants, particularly because we do not have the information we need to scientifically verify the model and explain the results and their consequences for the participants. I had understood from the OTAG meeting that we cannot adequately interpret the results and what they may mean (especially around the time of exposure and therefore the initial burden of exposure and so the subsequent potential health risks) until we have completed the study?

If you do find the wording of the ethics approval is still a concern, you may need to go back to the Ethics Committee to explain the methodology has changed and the wording of the approval may need to be revised to refer to advising participants (and the community and stakeholders) of the results of the study within two months of the results of analysis being available. I strongly recommend you take the Chair of OTAG with you to such a meeting so OTAG's advice can be provided and any queries answered.

If you and Jeff still wish to meet with Graeme (and presumably Bill/Fiona) I am happy to advise Graeme and get this arranged. If necessary, a further meeting between Graeme, Bill/Fiona, Don and John Hay can be arranged.

Kind regards - Sally

Sally Gilbert
Team Leader (Environment Team)
Environmental Health
Public Health Programmes
Public Health Directorate
Ministry of Health
DDI: s 9(2)(g)(ii)
Fax: 04 495 4401

<http://www.moh.govt.nz>
mailto:sally_gilbert@moh.govt.nz



Document Profile

PHP Physical Environment Filing (Archive

1996-2011)

Status:	Final	Drawer:	Toxic Substances
Date:	02/08/2004	Folder:	Organochlorines\Serum Dioxin
Title:	RE: Serum dioxin meeting 5 August 2004		Study\Phase 2\ESR
Author:	Sally Gilbert	File Location:	
Document Type:	Email	Unit:	
Summary:		Maintainer(s):	Annie Coughlan
Knowledge Content:			

Thank you for your email, David

There are a number of points of disagreement, resolution of which is probably best left for our meeting on Thursday...

Cheers - sally

"Phillips, David"
<David.Phillips@ESR.CRI.NZ>
02/08/04 01:43 pm
T "sally_gilbert@moh.govt.nz" <sally_gilbert@moh.govt.nz>
o
cc "Fowles, Jeff" <Jeff.Fowles@ESR.CRI.NZ>, "Swallow, Bill" <Bill.Swallow@esr.cri.nz>, "Thomson-Carter, Fiona" <Fiona.Thomson-Carter@esr.cri.nz>, graeme.gillespie@moh.govt.nz, fae_sidney@moh.govt.nz, Deborah_Read@moh.govt.nz, annie_coughlan@moh.govt.nz, peter_abernethy@moh.govt.nz
S RE: Serum dioxin meeting 5 August 2004
u
bj
e
ct

Dear Sally

Thank you for your e-mail regarding the Paritutu dioxin work, as per our recent discussions. As I mentioned we have given considerable thought to the blood test data and on reflection feel comment on the proposed course of action as recommended by OTAG and accepted by the MoH is merited. To enable us to proceed with all reasonable speed in any discussions, I thought it would be helpful to outline our thinking beforehand, namely:

1. The purpose of the study as stated in the tender document was "to collect and analyse samples of blood serum from selected past and current residents of Paritutu to ascertain possible exposure to dioxins, (foremostly 2,3,7,8 TCDD), related to the historical activities of Dow AgroSciences (formerly operating as IWD) in manufacturing 2,4,5-T at the Paritutu site....In particular, this is to assess the levels of dioxin (esp 2,3,7,8 TCDD) in the blood serum

of a group of Paritutu residents living in proximity to the IWD factory at suspected peak emission times, and therefore most likely to have been highly exposed. Comparisons will be made between this group's results and comparable strata from the 1998 MfE serum survey ."

2. The indicative sample size calculation, based on a required power and on anticipated differences between certain groups was 50. It was anticipated that this sample would enable us to observe differences of a certain [statistically significant] magnitude in various age groups. We were subsequently directed by yourselves to divide this in two parts; in the first instance targeting a smaller subgroup with the very highest probability of showing an effect should, there be any. The anticipation being at that time that it would require all 50 samples to show a significant effect.

3. The differences between observed and anticipated in the first set of samples were much bigger than anticipated such that the central research question has been answered, and as such, there would appear no justification to sample further residents.

4. The model developed has been validated, although with experience gained from this exercise, sensible modifications could be made for future use in similar situations. These observations in no way invalidate the model.

5. The results were, as you mention, discussed at OTAG and again, as you say, the advice to the Ministry was that the study be completed before any results are given to participants. We believe, for the reasons outlined above, the study is complete. The study, has, as we surmised, raised some interesting questions, however, these are ancillary to the central research question. For example, one of the questions you have highlighted refers to the timing of exposure. In the absence of emission and other substantive data further blood sampling would, we believe, be unlikely to give a more definitive answer to this question.

6. We would also mention that, as the report has highlighted, exposure was most likely due to emissions of a fugitive nature and as such it may be difficult/impossible to define one or a small number of particular time frames using blood data only. It would be a possibly unwise course of action to place significant faith in further testing as providing an unequivocal answer to any remaining questions about when and how much TCDD people were exposed to. At the moment, the potential peak body burdens can reasonably be expected to form a range spanning an adjustment of approximately two to four half-lives from current values. Further testing may or may not help clarify this range somewhat, but we do not believe this will result in any information of direct use for individuals.

7. Better understanding of the time of exposure will not enable us to give any more definitive information about individual or population health risks at this time, nor would it enable any intervention, as there is not one.

8. The course of action currently proposed raises, as I mentioned to you, a number of concerns, particularly in relation to our existing commitments and obligations, plus some significant challenges for further testing, in particular:

- A follow up study as proposed, involves a different objective and hence a different

purposive sampling design and participant selection criteria. It would be potentially misleading for the results from the two studies to be combined.

- The supporting documents distributed to participants [developed in conjunction with yourselves] and which were part of the submission for ethical approval, stated that individual results will be available 'up to two months' from the taking of blood.
- Any significant delay in distributing the results could have a number of undesirable effects, including raising concerns among the local community, which could impact on recruitment for further testing.
- Similarly, ESR has worked hard to establish trust and gain acceptance from all parts of the Paritutu community. The potential for accusations of cover up, adverse press publicity, etc in the proposed course of action are considerable and disproportionate to the possible gains from the course of action. We also believe the associated risks to ESR are significant.

9. The report has been externally [internationally] peer reviewed, positively, and some minor changes have been made and it is ready for distribution. It would seem logical and consistent with stated commitments to make the information available as originally agreed, and then to initiate the work as recently proposed.

10. While we understand that the results will raise some 'difficult' questions, we believe this can largely be addressed by communicating the ongoing nature of the investigation, and emphasizing the Ministry's commitment to providing as much clarity as possible on residual issues within a reasonable timeframe. This is not, as we know, a new issue confronting the Paritutu community.

11. We can foresee no 'downside' to such a course of action, whereas to proceed with further testing without disseminating the existing findings could, as indicated above, produce major 'challenges'.

12. We hope this explains our view of the issues and options somewhat better, and we hope can agree a way forward. We agree with you, that, depending on the outcome of our discussions, it may be important to explore other perspectives, especially relating to the existing ethics committee approval and the information supplied to participants.

Fiona, Jeff and I have commitments elsewhere on Thursday afternoon but would be happy to discuss these issues further on Thursday morning. The meeting between Graeme Gillespie and Bill Swallow is, I understand, timed to finish at 11.30 and Bill has indicated to me he would be happy to join the discussion after that. One hour should, we feel, be more than enough time.

If any of the above requires clarification please feel free to contact me.

Kind regards

David